

Remote and Hybrid Work Arrangements: Effects on Organizational Productivity, Employee Well-Being, and Managerial Practice

Evidence from a Longitudinal Panel Study Across 218 Organizations (2020–2023)

Dr. Chiara Benedetti

Marcus Tran

Dr. Oluwaseun Adeyemi

Department of Organizational Behavior, London Business School

INSEAD Social Innovation Centre

Lagos Business School, Pan-Atlantic University

c.benedetti@london.edu

Abstract

The COVID-19 pandemic precipitated the largest unplanned experiment in remote work in modern economic history, creating a natural opportunity to examine the effects of work arrangement on organizational outcomes that had previously been studied only in limited contexts. This paper reports results from a four-wave longitudinal panel study tracking 218 organizations across 11 industries and 14 countries from Q1 2020 to Q4 2023. We measure productivity through objective output metrics (revenue per employee, units produced, project completion rates) and managerial assessments, and well-being through validated scales (PHQ-9, GAD-7, Work-Life Balance Scale). Our findings reveal a consistent inverted-U relationship between remote work intensity and productivity, with peak performance observed at 2–3 remote days per week (hybrid arrangements). Fully remote organizations show a 7.3% average productivity deficit relative to hybrid, concentrated in roles requiring high task interdependence. Conversely, well-being scores are positively associated with remote intensity, with each additional remote day associated with a 0.18-point improvement on a 10-point well-being scale ($p < 0.001$). We discuss implications for hybrid work policy design, managerial capability development, and the distributional effects of work arrangement flexibility across demographic groups.

Keywords: remote work, hybrid work, organizational productivity, employee well-being, work-life balance, managerial practice

1. Introduction

1.1 Context and Motivation

Prior to 2020, the academic literature on remote work was primarily characterized by single-organization case studies and cross-sectional surveys, constrained by self-selection effects and limited generalizability. The pandemic-imposed shift to remote work eliminated the

self-selection problem for the initial transition and created longitudinal variation in remote work intensity as organizations moved through full-remote, return-to-office, and hybrid phases at different speeds and with different policies. This natural experiment, while subject to its own confounds, provides an unprecedented dataset for examining the productivity-flexibility tradeoff.

1.2 Research Questions

This study addresses four primary questions: (i) What is the net effect of remote and hybrid work on organizational productivity, controlling for industry and role type? (ii) How do well-being outcomes vary with remote work intensity? (iii) What organizational and managerial practices moderate the productivity effects of remote work? (iv) Are the effects of remote work arrangement distributed equally across gender, caregiving status, and seniority?

2. Data and Methods

2.1 Sample

Participating organizations were recruited through industry partnerships and voluntary enrollment, with a deliberate oversampling of knowledge-intensive sectors (technology, finance, consulting, professional services) where remote work is most feasible and most extensively adopted. Final sample: 218 organizations, 47,312 individual employees, 11 industries, 14 countries across North America, Western Europe, East Asia, and Sub-Saharan Africa. Longitudinal attrition was 14.3% across four waves, with attrition tests showing no significant differences from retained sample on baseline characteristics.

2.2 Measurement

Productivity: Primary measures were objective output data (revenue per employee, project completion rate, quality-adjusted units produced) supplemented by direct managerial assessments using a validated 12-item scale. Well-being: Employee surveys administered at each wave included the PHQ-9 (depression), GAD-7 (anxiety), the Work-Life Balance Scale (Hayman, 2005), and a single-item life satisfaction measure. Remote work intensity was measured as self-reported percentage of working days spent remote, validated against HR system records where available.

3. Results

3.1 Productivity Effects

Fixed-effects panel regression models, controlling for organization size, industry, country, and time trends, show a statistically significant inverted-U relationship between remote work intensity and productivity. Organizations with 40–60% remote work intensity (roughly 2–3 days per week) outperformed both fully in-office (productivity index: 100) and fully remote arrangements (productivity index: 92.7) by an average of 4.8 percentage points. Heterogeneity analysis reveals that the productivity penalty for full remote is concentrated in roles requiring high task interdependence (coefficient: -11.2, $p < 0.001$), creative collaboration (coefficient: -8.7, $p < 0.01$), and on-the-job mentoring (coefficient: -9.3, $p < 0.01$), while roles with high autonomy and low interdependence show no significant productivity difference across work arrangements.

3.2 Well-Being Outcomes

Well-being scores showed a positive linear association with remote work intensity across all four waves. Each additional remote day per week was associated with a 0.18-point improvement in composite well-being (95% CI: 0.14–0.22, $p < 0.001$), driven primarily by improved work-life balance scores and reduced commute-related stress. Notably, the well-being benefits of remote work were smaller for employees without dedicated home office space (interaction coefficient: -0.11) and for primary caregivers of young children, who showed higher anxiety scores under full-remote arrangements despite higher average work-life balance ratings.

3.3 Demographic Heterogeneity

The productivity benefits of hybrid arrangements were distributed relatively evenly across demographic groups. However, women in senior roles showed significantly larger well-being improvements from hybrid work (mean additional benefit: 0.31 points, $p < 0.05$) relative to men, consistent with the disproportionate caregiving burden documented in this group. Conversely, early-career employees (under 5 years of experience) showed steeper productivity declines under full-remote arrangements (coefficient: -14.3) than senior employees (coefficient: -4.8), underscoring the importance of in-person mentoring and socialization for professional development.

4. Discussion

4.1 Managerial Practice Moderators

Organizations with structured hybrid policies — specifying anchor days for team in-person attendance, explicit norms for synchronous vs. asynchronous communication, and training for managers in remote team leadership — showed substantially smaller productivity penalties under high remote intensity (coefficient: -2.1 vs. -9.8 for organizations without structured policies). This finding suggests that the productivity costs of remote work are substantially organizational in origin and are amenable to managerial intervention, rather than being inherent to the remote modality.

4.2 Implications for Policy Design

The evidence presented here supports a default hybrid policy of 2–3 office days per week for interdependent knowledge work, with flexibility for more remote-intensive arrangements in autonomous, output-measurable roles. One-size-fits-all return-to-office mandates appear suboptimal from both a productivity and retention perspective, given the strong well-being benefits of flexibility and documented effects on employee turnover intent. Organizations should invest in manager capability development, physical workspace redesign to prioritize collaboration over individual work (which employees can do at home), and equity-conscious flexibility policies that account for the differential constraints faced by caregivers and early-career employees.

5. Conclusions

This longitudinal panel study provides robust evidence that hybrid work arrangements — particularly those structured around 2–3 in-person days per week — optimize the productivity-well-being tradeoff relative to both fully in-office and fully remote alternatives. The productivity costs of full

remote work are real but concentrated in interdependent roles and substantially modifiable through deliberate organizational design. The well-being benefits of flexibility are large, consistent, and distributed broadly, though with important heterogeneity by caregiving status and career stage. Organizations seeking to design effective hybrid policies should move beyond binary in-office/remote debates toward nuanced, role-differentiated frameworks supported by genuine managerial capability and equitable implementation.

References

- [1] Bloom, N., Liang, J., Roberts, J., & Ying, Z. J. (2015). Does working from home work? Evidence from a Chinese experiment. *Quarterly Journal of Economics*, 130(1), 165–218.
- [2] Barrero, J. M., Bloom, N., & Davis, S. J. (2023). The evolution of working from home. *Journal of Economic Perspectives*, 37(4), 23–49.
- [3] Kroenke, K., Spitzer, R. L., & Williams, J. B. W. (2001). The PHQ-9: Validity of a brief depression severity measure. *Journal of General Internal Medicine*, 16(9), 606–613.
- [4] Hayman, J. (2005). Psychometric assessment of an instrument designed to measure work life balance. *Research and Practice in Human Resource Management*, 13(1), 85–91.
- [5] Gibbs, M., Mengel, F., & Siemroth, C. (2021). Work from home and productivity: Evidence from personnel and analytics data on information workers. *Journal of Political Economy Microeconomics*, 1(1), 7–41.
- [6] Yang, L., et al. (2022). The effects of remote work on collaboration among information workers. *Nature Human Behaviour*, 6(1), 43–54.
- [7] McKinsey Global Institute. (2023). *Generative AI and the future of work in America*. McKinsey & Company.